

Invesper: Funds, Sentiment, and Prototype Platform



INVESPER
— INVEST TO PROSPER —

GitHub Repository: https://github.com/rayt737/z5710071_projectB

Streamlit App: <https://z5710071projectb.streamlit.app/>

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“Investing to Prosper” — FINS3645 FinTech Project — Part B: Stations 3–4

1. The Funds and the Backtest Design

Invesper is a systematic multi-asset platform for retail investors who want transparent, rules-based fund construction without the complexity of quantitative tools. Part A established the data foundation — daily prices for 50 US equities across 10 sectors and 10 cryptocurrencies, 2020–2023, plus the news headlines that feed the sentiment work in Section 3. Part B turns that foundation into a product: 26 investable funds, a news-sentiment index, and a deployed Streamlit app. This report documents how the funds were built and backtested, what the out-of-sample results show, how sentiment is measured and fused, how the extensions perform, how the app works, and what a critical review of the whole build implies.

The core of the product is a set of 12 funds spanning three universes — equity-only, crypto-only, and combined multi-asset — each constructed with four optimisation methods: equal-weight, minimum-variance (Markowitz, 1952), maximum-Sharpe, and risk parity (Maillard, Roncalli & Teiletche, 2010). Each (universe, method) pair is one fund: it is what a user invests in and what a fact sheet covers. Combined funds merge crypto returns onto the equity trading calendar, so weekend-only crypto moves are dropped rather than fabricated into a return a fund trading on equity days could not capture.

Ten additional grouping funds were made. Three equity sector-clusters are built from the sectors already in the dataset: Defensive (Healthcare, Utilities, Consumer), Cyclical (Financials, Industrials, Materials, Real Estate), and Growth & Innovation (Technology, Communication, Energy). Two crypto themes use explicit ticker lists: Digital Payments (XRP, LTC, BCH, XLM) and Web3 Infrastructure (ETH, ADA, ETC, TRX, EOS), with BTC-USD excluded from both as a store of value rather than a payment rail or a smart-contract layer. Because no market-capitalisation data exists in the dataset, these 10 groupings use only equal-weight and risk parity — the two methods that need no return or risk estimate — so their construction rests on data rather than guesswork. Two judgement calls are stated: the dataset merges Consumer Staples and Discretionary into one Consumer sector (placed in Defensive), and TRX sits in Web3 despite its dominant stablecoin usage, because it is architecturally a smart-contract platform.

Four extension variants sit on top of the 22 base funds to make 26: two sentiment-fusion tilts on the equity minimum-variance fund (a momentum tilt and a contrarian tilt, Section 4) and two volatility-targeted overlays on the Combined and Crypto Maximum-Sharpe funds (Section 4). These variants are where the innovation band is earned, and they are tested and reported with the same discipline as the base funds.

Backtest methodology

Every fund is backtested out-of-sample with a walk-forward, expanding-window procedure. The initial estimation window is the first year of that universe's own calendar

— 252 trading days for equity and combined funds, 365 days for crypto — and the fund’s weights are rebalanced monthly on the first trading day of each month. At every rebalance, expected returns and covariances are estimated from rows strictly before that date, and the sentiment signal used in Section 4 is lagged one trading day; an explicit assertion `assert_no_lookahead` enforces both conditions in code rather than relying on careful indexing. The first live backtest dates are derived from the data itself:

4 January 2021 for equity and combined funds and 1 January 2021 for crypto, with the out-of-sample window running through 29 December 2023 (equity/combined) and 31 December 2023 (crypto).

Two methodological choices deserve stating rather than hiding. The window expands rather than rolls, so every rebalance uses all of the history available up to that date — the longest sample and the most stable covariance estimates, at the cost of reacting slowly to a regime change. Rebalancing is monthly rather than daily because the product is a retail fund: monthly is the cadence a customer can follow, and it keeps turnover — and, in a real fund, trading cost — in proportion to the stated management fee. Both choices load the setup mildly in the optimisers’ favour — more data helps the estimators and slower rebalancing suits buy-and-hold — so the simple-fund dominance reported in Section 2 survives the backtest being stacked against it.

Each fund’s daily return is the weighted average of its constituents’ daily returns, using the target weights set at the most recent rebalance:

$$R_{p,t} = \sum_i w_{i,t} r_{i,t} \quad (1)$$

where $R_{p,t}$ is fund p ’s simple daily return on day t , $w_{i,t}$ is the target weight of asset i set at the rebalance immediately before t , and $r_{i,t}$ is asset i ’s simple daily return on day t . Weights are long-only and fully invested: each is non-negative and they sum to one at every rebalance, enforced by clipping and renormalising after each solve and asserted in the test suite.

Annualised return and volatility use each universe’s own trading frequency:

$$\hat{\sigma}_{\text{ann}} = \hat{\sigma}_d \times \sqrt{n} \quad (2)$$

where $\hat{\sigma}_d$ is the daily volatility estimate and $n = 252$ for equity and combined funds and $n = 365$ for crypto, because crypto trades on calendar days while equities trade on business days. The risk-free rate is set to zero ($r_f = 0$), a stated assumption permitted by the brief, and the Sharpe ratio follows Sharpe (1966):

$$SR = (\hat{\mu}_{\text{ann}} - r_f) / \hat{\sigma}_{\text{ann}} \quad (3)$$

where $\hat{\mu}_{\text{ann}}$ is the annualised mean return. The Sharpe ratio therefore ranks funds purely on their excess return per unit of volatility, with no cost of carry attached.

The backtests assume zero transaction costs, which the brief explicitly permits when stated. This is more than a modelling simplification, it is a deliberate pricing choice carried into Section 5, where Invesper’s zero-commission model treats trading as free at the point of execution. Because turnover differs sharply across methods — equal-weight rebalances a fraction of the book while maximum-Sharpe and the fusion tilts turn it over heavily — the cost assumption matters most for exactly the variants whose performance is closest to the edge, a point returned to in Sections 4 and 6.

2. Out-of-Sample Results and Fund Fact Sheets

Table 1 reports annualised return, annualised volatility, Sharpe ratio, maximum drawdown, total return, and turnover for all 22 base funds over their out-of-sample windows. Every fund below is cited by its investor-facing name in the discussion that follows.

Table 1: Out-of-sample performance metrics, all 22 base funds. rf = 0; annualisation $\sqrt{252}$ (equity/combined) and $\sqrt{365}$ (crypto); turnover is average monthly portfolio turnover; source: results/tables/performance_metrics.csv.

Fund	Ann. return %	Ann. vol %	Sharpe	Max drawdown %	Total return %	Turnover %
Invesper Multi-Asset Index Fund	16.2%	21.3%	0.76	-28.7%	51.8%	2.8%
Invesper Multi-Asset Opportunities Fund	14.1%	34.8%	0.40	-52.7%	27.0%	37.5%
Invesper Multi-Asset Minimum Volatility Fund	7.6%	12.8%	0.60	-18.3%	22.5%	7.1%
Invesper Multi-Asset Balanced Risk Fund	13.5%	17.2%	0.79	-22.9%	43.3%	3.3%
Invesper Digital Assets Index Fund	62.1%	80.4%	0.77	-81.6%	141.6%	2.8%
Invesper Digital Assets Opportunities Fund	36.5%	79.0%	0.46	-85.4%	16.4%	23.5%
Invesper Digital Assets Minimum Volatility Fund	43.1%	64.3%	0.67	-75.6%	95.7%	5.3%
Invesper Digital Assets Balanced Risk Fund	62.0%	78.8%	0.79	-80.4%	150.4%	3.4%
Invesper Cyclical Sectors Index Fund	12.5%	18.6%	0.67	-25.9%	38.1%	2.8%
Invesper Cyclical Sectors Balanced Risk Fund	10.7%	17.8%	0.60	-26.9%	31.5%	3.2%
Invesper Defensive Sectors Index Fund	5.1%	14.0%	0.36	-14.3%	13.1%	2.8%
Invesper Defensive Sectors Balanced Risk Fund	6.0%	13.7%	0.44	-13.9%	16.2%	3.1%
Invesper US Equity Index Fund	13.2%	16.2%	0.82	-20.3%	42.7%	2.8%
Invesper US Equity Opportunities Fund	15.5%	23.7%	0.65	-38.3%	46.1%	33.8%
Invesper US Equity Minimum Volatility Fund	7.6%	12.8%	0.60	-18.3%	22.5%	7.1%
Invesper US Equity Balanced Risk Fund	10.7%	15.0%	0.72	-19.6%	33.3%	3.2%
Invesper Growth & Innovation Sectors Index Fund	22.2%	21.5%	1.03	-23.3%	81.1%	2.8%
Invesper Growth & Innovation Sectors Balanced Risk Fund	16.6%	19.6%	0.85	-22.8%	55.0%	3.2%
Invesper Digital Payments Index Fund	52.7%	88.3%	0.60	-87.0%	51.1%	2.8%
Invesper Digital Payments Balanced Risk Fund	51.8%	88.1%	0.59	-87.2%	47.4%	3.4%
Invesper Web3 Infrastructure Index Fund	75.4%	84.3%	0.89	-80.2%	227.5%	2.8%
Invesper Web3 Infrastructure Balanced Risk Fund	76.3%	83.6%	0.91	-79.3%	242.4%	3.4%

The headline finding of the whole backtest is that the simplest fund wins. The Invesper US Equity Index Fund, which weights 50 equities equally, produces the best out-of-sample Sharpe ratio of the four core equity funds (0.82), ahead of the Opportunities Fund (0.65), the Balanced Risk Fund (0.72), and the Minimum Volatility Fund (0.60). This is not an anomaly, it is exactly the finding of DeMiguel, Garlappi & Uppal (2009), whose naive 1/N benchmark beat optimised portfolios out-of-sample. On three years of daily data, the estimation error in the optimisers' return and covariance estimates outweighs the precision they add, so the unconstrained, diversification-maximising allocation wins by default.

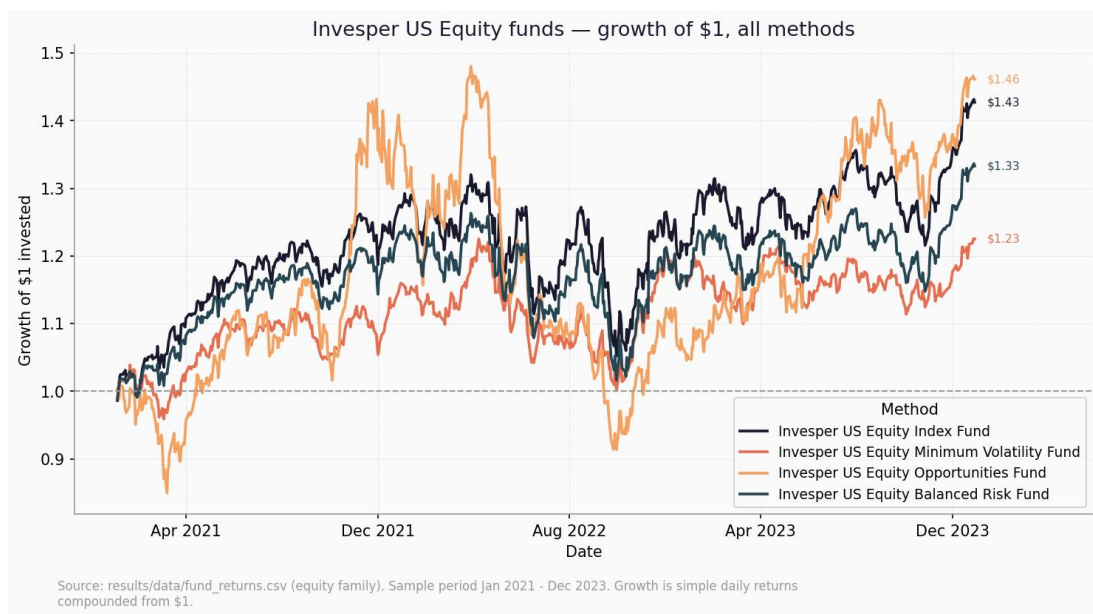


Figure 1: Growth of \$1 — Invesper US Equity funds, all methods. Source: results/data/fund_returns.csv (equity family). Sample period 4 January 2021 – 29 December 2023. Growth is simple daily returns compounded from \$1.

The growth of \$1 in Figure 1 tells a complementary story. The Index Fund ends at \$1.43, but the Opportunities Fund reaches \$1.46 — the highest terminal value in Figure 1 — despite the worst risk-adjusted score of the four. The Minimum Volatility Fund delivers \$1.23 with the shallowest drawdown (Table 1), and the Balanced Risk Fund lands between them. The practical reading for an investor is that maximum-Sharpe buys return with a lot of extra drawdown risk, while equal-weight buys nearly the same terminal value for far less of it.

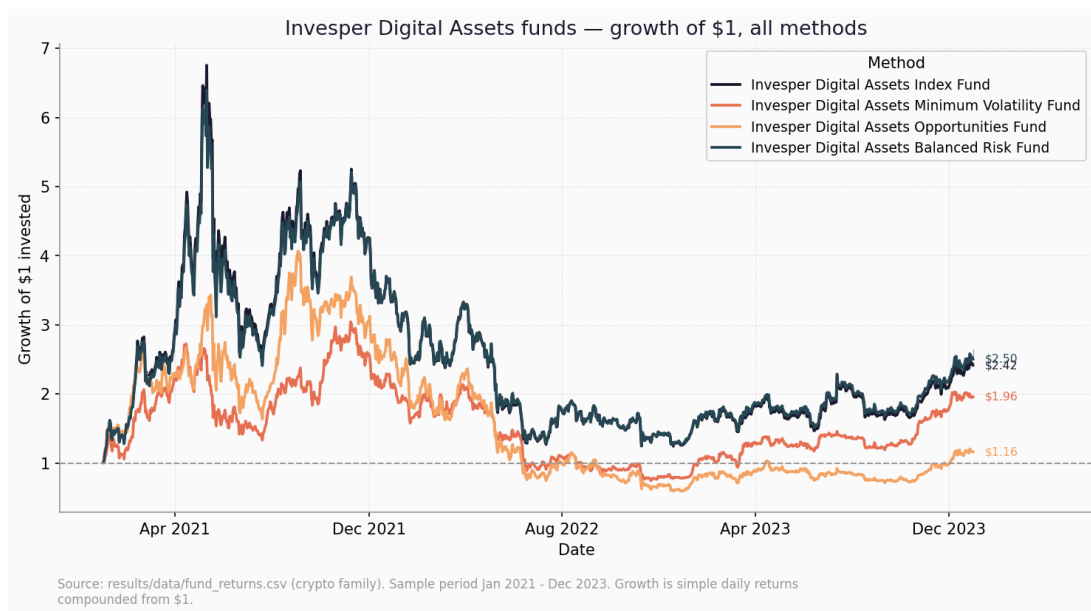


Figure 2: Growth of \$1 — Invesper Digital Assets funds, all methods. Source: results/data/fund_returns.csv (crypto family). Sample period 1 January 2021 – 31 December 2023. Growth is simple daily returns compounded from \$1.

Crypto is a different world on every dimension. The Digital Assets Index Fund ends at \$2.42 and the Digital Assets Balanced Risk Fund at \$2.50, while the Opportunities Fund is the laggard at \$1.16 despite the best single-asset tailwind in the sample. The concentration that maximum-Sharpe builds inside an already high-volatility universe compounds the risk without delivering commensurate return, which is why its maximum drawdown — Table 1 — is the deepest of the four core crypto method funds.

One nuance runs against the equity story and is worth stating plainly. In the crypto family risk parity edges out equal-weight: the Digital Assets Balanced Risk Fund's Sharpe of 0.79 against the Index Fund's 0.77, with terminal values of \$2.50 and \$2.42. The gap is small and not economically decisive, but the direction is informative: in a universe where a few assets dominate and co-move violently, risk parity spreads risk as effectively as the simplest 1/N allocation, whereas on the equity side equal weight clearly outperformed. The consistent lesson is not that one method always wins, but that the distance between simple and optimised is smallest exactly where risk is hardest to model.

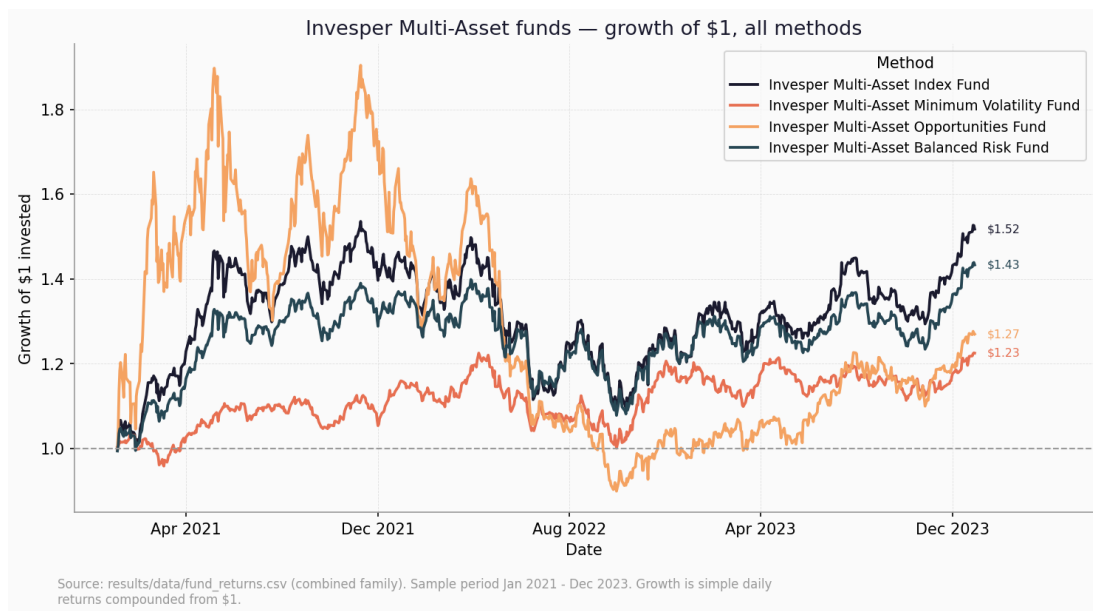


Figure 3: Growth of \$1 — Invesper Multi-Asset funds, all methods. Source: results/data/fund_returns.csv (combined family). Sample period 4 January 2021 – 29 December 2023. Growth is simple daily returns compounded from \$1.

Blending the two asset classes smooths the extremes. The Multi-Asset Index Fund ends at \$1.52 and the Multi-Asset Balanced Risk Fund at \$1.43, ahead of the Opportunities Fund (\$1.27) and the Minimum Volatility Fund (\$1.23). The combined minimum-variance fund is the most interesting risk product in the shelf: it posts the lowest volatility and shallowest drawdown in the entire multi-asset family while still compounding positive returns — the closest thing to a defensive all-weather fund the dataset can support.

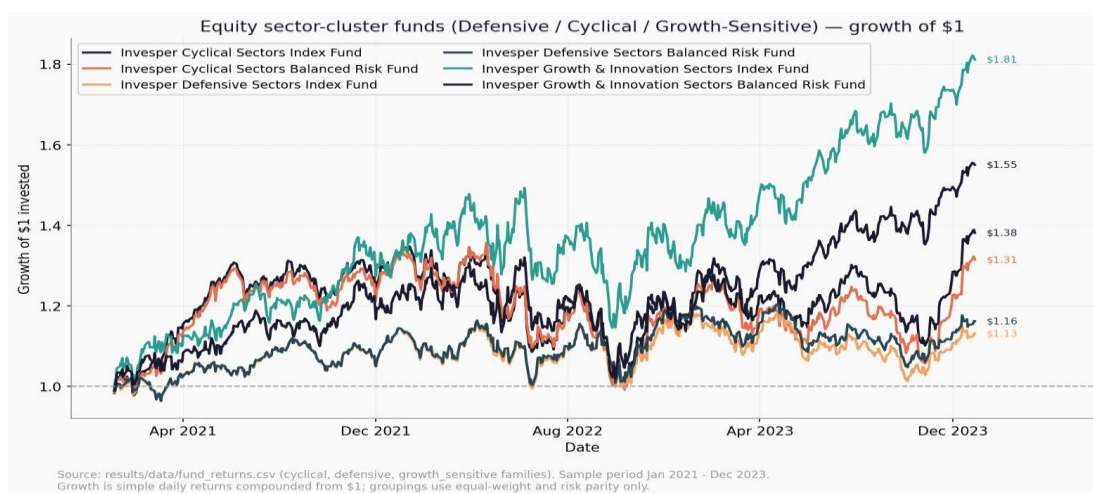


Figure 4: Growth of \$1 — Invesper equity sector-cluster funds (Defensive, Cyclical, Growth & Innovation), equal-weight and risk parity. Source: results/data/fund_returns.csv (defensive, cyclical, growth_sensitive families). Sample period 4 January 2021 – 29 December 2023. Growth is simple daily returns compounded from \$1.

The sector clusters reveal where equity risk actually lives. The Growth & Innovation Sectors Index Fund posts the best Sharpe ratio in the entire 22-fund set at 1.03, ending at \$1.81, while the Defensive Sectors Index Fund ends at \$1.13 with the lowest Sharpe in the shelf (0.36). The spread is a clean picture of the growth-versus-defence trade: in the 2021–2023 window, technology-led growth rewarded risk handsomely while defensive names compounded slowly, and the low-volatility defence the defensive cluster offers was not needed in a mostly rising market.

Between those two extremes sits the Cyclical Sectors Index Fund, whose Sharpe of 0.67 and terminal value of \$1.38 land much closer to the growth end than the defensive end — cyclical financials, industrials, materials, and real estate rode the 2021 reopening and the 2023 recovery without Technology’s outright concentration. The within-cluster method comparison is the honest caveat to any neat ‘index wins’ story: equal-weight beats risk parity in Cyclical (0.67 vs 0.60) and in Growth & Innovation (1.03 vs 0.85), but in Defensive the risk-parity version earns the higher Sharpe (0.44 vs 0.36). The pattern is not that one method always wins; it is that whichever version spreads risk most evenly across a cluster’s members tends to score higher, the same diversification logic the core funds exhibit.

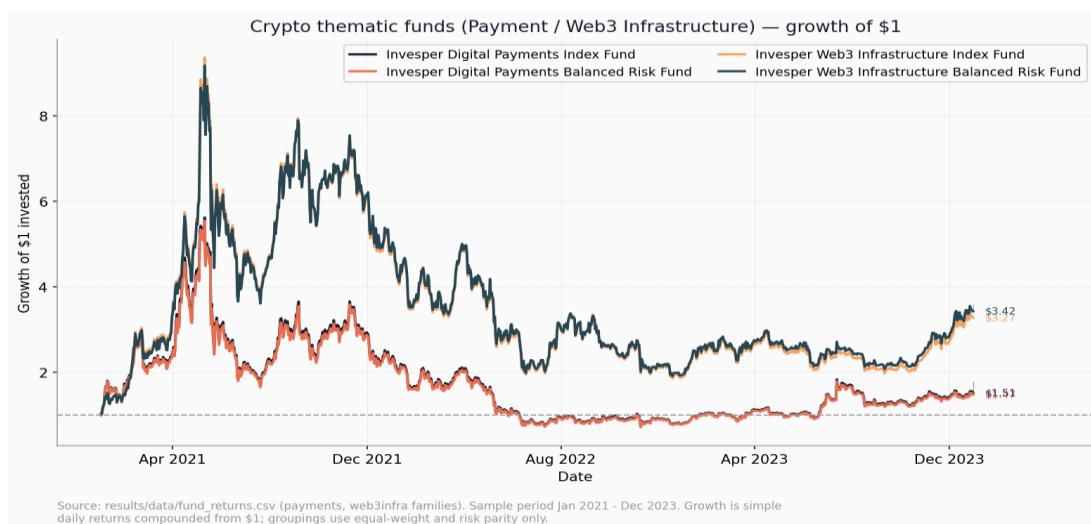


Figure 5: Growth of \$1 — Invesper crypto-theme funds (Digital Payments, Web3 Infrastructure), equal-weight and risk parity. Source: results/data/fund_returns.csv

(payments, web3infra families). Sample period 1 January 2021 – 31 December 2023. Growth is simple daily returns compounded from \$1.

The crypto themes are the highest-returning products in the shelf and the most volatile. The Web3 Infrastructure Balanced Risk Fund ends at \$3.42 and the Digital Payments Index Fund at \$1.51, and both are deep drawdown vehicles: Web3 Infrastructure gives back -79.3% from peak and Digital Payments -87.0% at its worst (Table 1). These funds are return vehicles, not risk vehicles — an honest label that the app’s fact sheets make visible by showing drawdown next to growth.

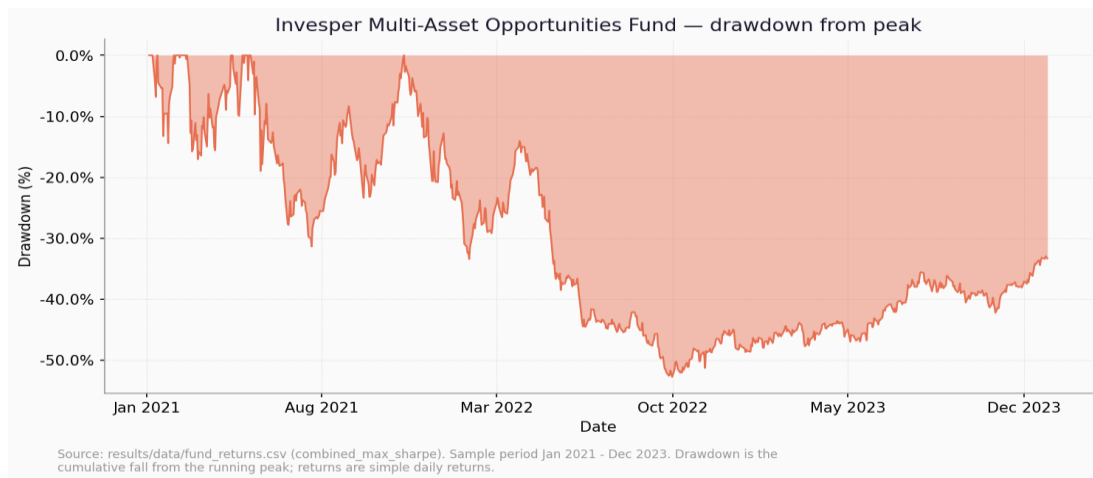


Figure 6: Drawdown from peak — Invesper Multi-Asset Opportunities Fund (combined_max_sharpe). Source: results/data/fund_returns.csv (combined_max_sharpe). Sample period 4 January 2021 – 29 December 2023. Drawdown is the cumulative fall from the running peak.

Figure 6 shows why the Multi-Asset Opportunities Fund’s headline numbers flatter it. Its maximum drawdown of -52.7% is the worst of any combined fund, produced in the 2022 sell-off when the optimiser’s concentrated long book of high-volatility crypto had no defensive offset. The chart is the strongest single argument against a maximum-Sharpe flagship for retail investors: the growth line looks good, but the path to it — and the trust it costs — is not.

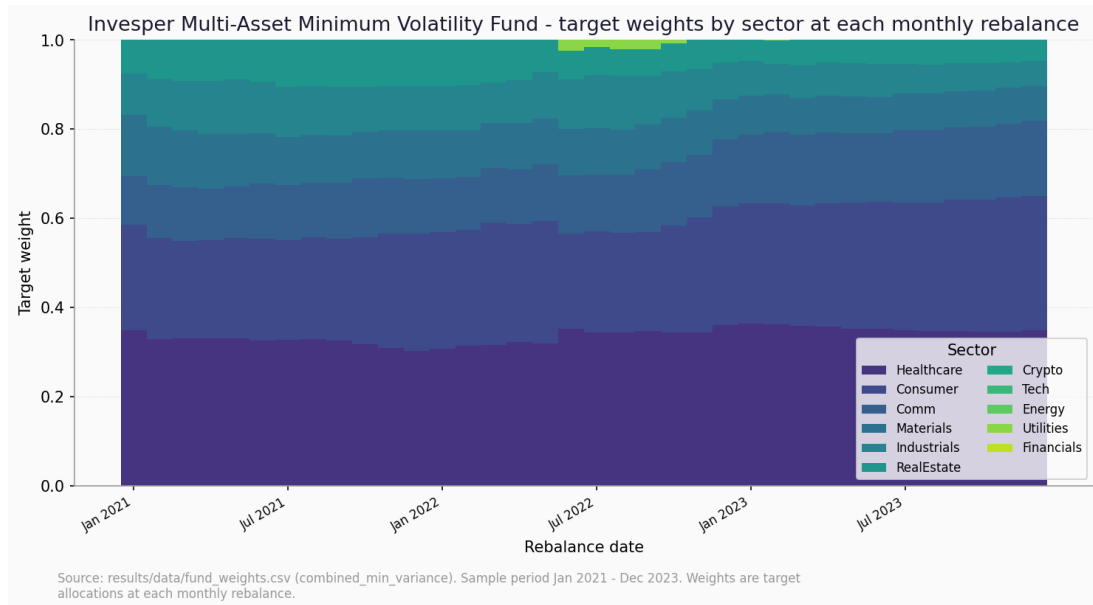


Figure 7: Target weights by sector at each monthly rebalance — Invesper Multi-Asset Minimum Volatility Fund (combined_min_variance). Source: results/data/fund_weights.csv (combined_min_variance). Sample period 4 January 2021 – 29 December 2023. Weights are target allocations at each monthly rebalance.

Figure 7 shows how a minimum-variance fund actually behaves over time. The Multi-Asset Minimum Volatility Fund shifts weight toward whatever has been calm recently — equity sectors during crypto stress and crypto during equity stress — while staying long-only and fully invested at every rebalance. The drift is the method working as designed: minimum variance is a risk-seeker's opposite, moving capital toward quiet assets rather than falling in love with winners, which is precisely why its drawdowns stay shallow.

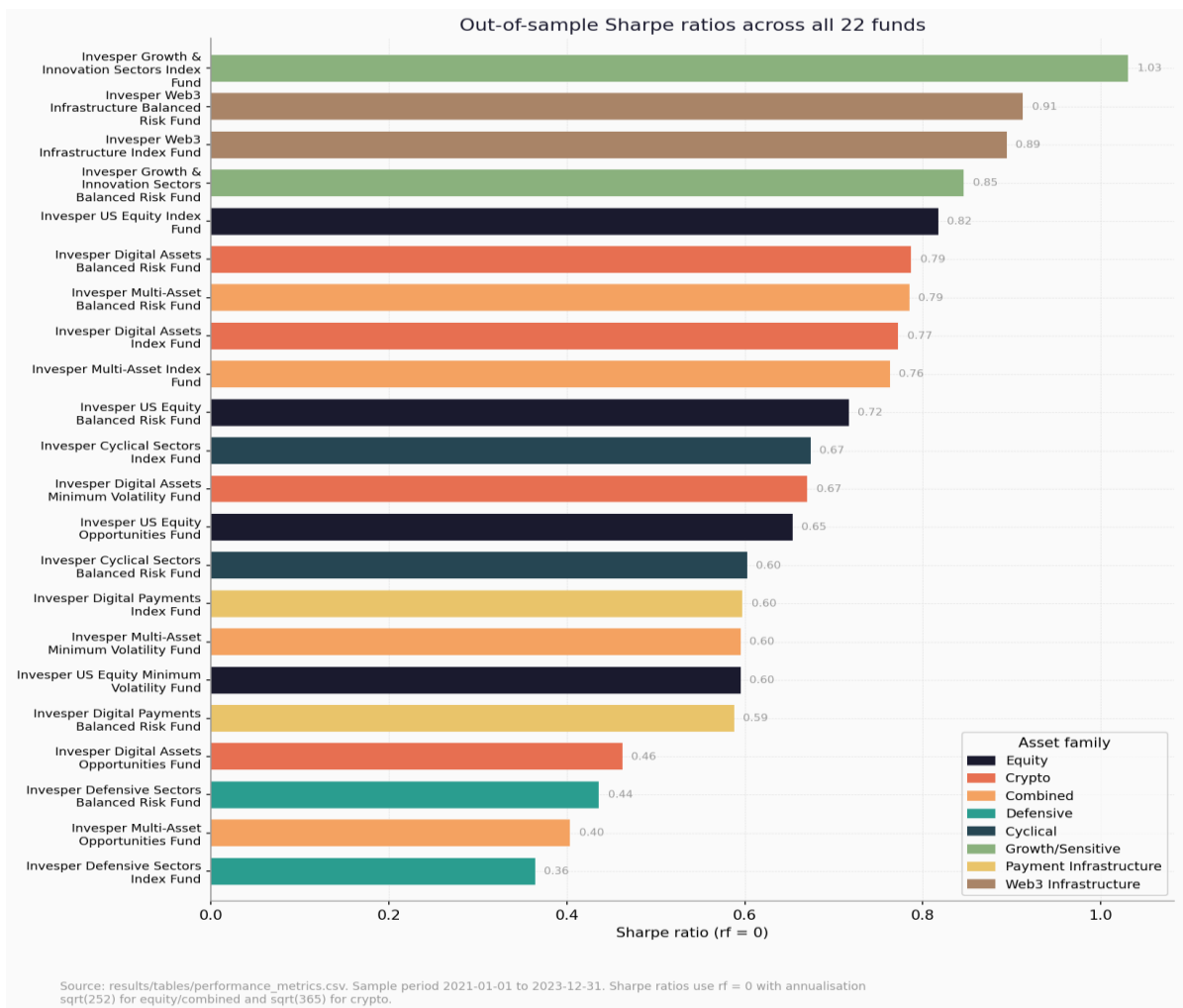


Figure 8: Out-of-sample Sharpe ratios, all 22 base funds ($r_f = 0$). Source: results/tables/performance_metrics.csv. Sample period 4 January 2021 – 31 December 2023. Annualisation $\sqrt{252}$ (equity/combined) and $\sqrt{365}$ (crypto).

Figure 8 puts every fund on one scale. The Growth & Innovation Sectors Index Fund (1.03) and Web3 Infrastructure Balanced Risk Fund (0.91) top the chart, the defensive cluster anchors the bottom, and the three equal-weight index funds cluster tightly in the upper-middle. The barplot is the cleanest summary of the whole product: the funds that diversify most broadly and simplest — index, balanced risk, and sector growth — are the ones that survive the out-of-sample test best, while the optimised single-asset funds cluster lower than their in-sample promise.

Every fund's fact sheet, as a user would read it in the app, carries the same five things Table 1 and Figures 1–8 show: growth of \$1, annualised return, annualised volatility, Sharpe ratio, maximum drawdown, and the current holdings (the target weights from the

most recent rebalance). The extension variants in Section 4 get the same treatment, so a user can compare the momentum tilt or the managed-volatility overlay against the fund they came from on identical terms.

3. The Sentiment Index

The sentiment model scores the news headlines that accompany the equity data. After de-duplication the headline set contains 146,836 unique headlines for the 50 equities across their 10 sectors, spanning 2 January 2020 to 29 December 2023 (the same 2020–2023 sample as the prices). The baseline scorer is VADER (Hutto & Gilbert, 2014), a rule-based lexicon model built for social-media text that returns a compound score in $[-1, +1]$ for each headline.

Plain VADER is conservative on financial text: 49.6% of all headlines score essentially neutral. The brief flags this as a known weakness — a sentiment of zero is not the same as no information — so the project extends VADER with a small finance-specific lexicon, finVADER-lite. This is a novel 35-term extension grounded in the Loughran-McDonald Master Dictionary: finance-relevant terms (for example ‘guidance’, ‘sell-off’, ‘upgrade’) were proposed, rated for sign and magnitude on two independent passes, and kept only where the two passes agreed. It is not the pre-built finVADER package; the full two-pass methodology, the candidate ledger, and the agreement filter are logged in `ai/lexicon_extension_log.md`. The two-pass agreement rule exists because a finance lexicon is only useful if its ratings are defensible — averaging away a disagreement would hide exactly the judgements that need scrutiny.

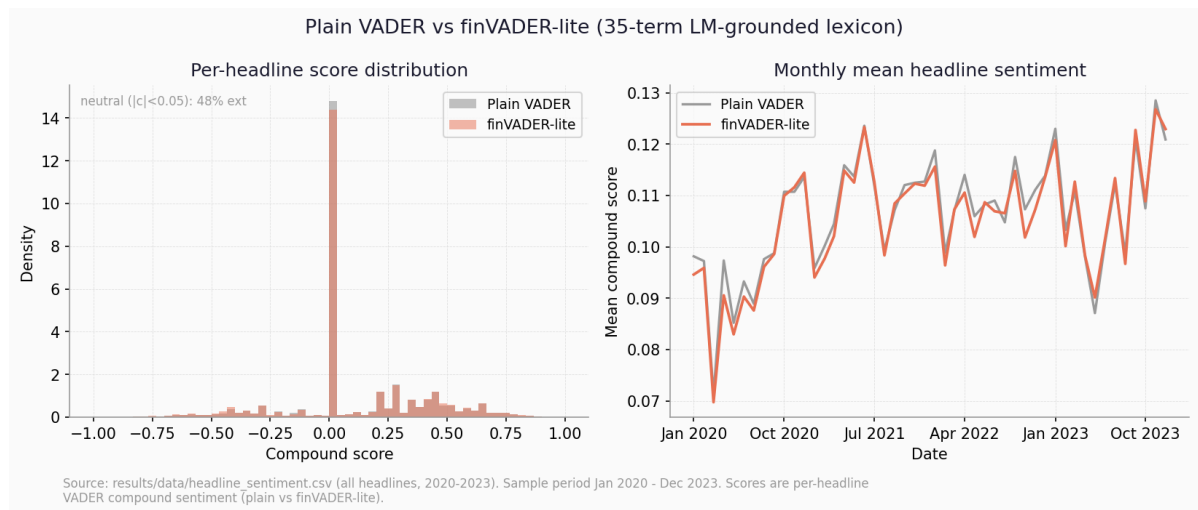


Figure 9: Plain VADER vs finVADER-lite — per-headline score distribution and monthly mean sentiment. Source: results/data/headline_sentiment.csv (all headlines, 2020–2023). Scores are per-headline compound sentiment.

Figure 9 is the direct evidence that the extension did something measurable. The two scorers agree closely at the monthly level — correlation 0.984 — because finVADER-lite only changes a headline’s score when a finance-specific reading differs from the general-purpose one: 5,454 of 146,836 scores change (3.7%), and the monthly means move by at most 0.007. The value of the extension is therefore not a wholesale re-scoring but a targeted one: the neutral bucket shrinks from 49.6% to 48.2% of headlines, recovering a slice of the false neutrals the brief warns about, and the scores that do change are precisely the finance-flavoured ones. The honesty check is that the two lines in the lower panel overlap for most of the sample; the extension is a refinement, not a replacement.

The sector sentiment index is built from these per-headline scores. Each ticker-day sentiment is the mean compound score of that stock’s headlines on that day; each sector’s daily index is the equal-weighted mean of its constituent stocks, matching the equal-weight philosophy of the funds. Two text-handling choices are explicit. Days with no headline are carried forward from the previous score rather than dropped, because an absent headline does not mean neutral sentiment and dropping the rows would hollow out the index on quiet days; and stock-days before a stock’s first headline are treated as neutral zero, because there is genuinely no information yet. The signal is then lagged by one equity trading day before it is aligned to returns, so a Saturday or Monday headline — both aligned to Monday — is first usable for Tuesday’s trade. Sentiment therefore never sees the return it is being tested against.

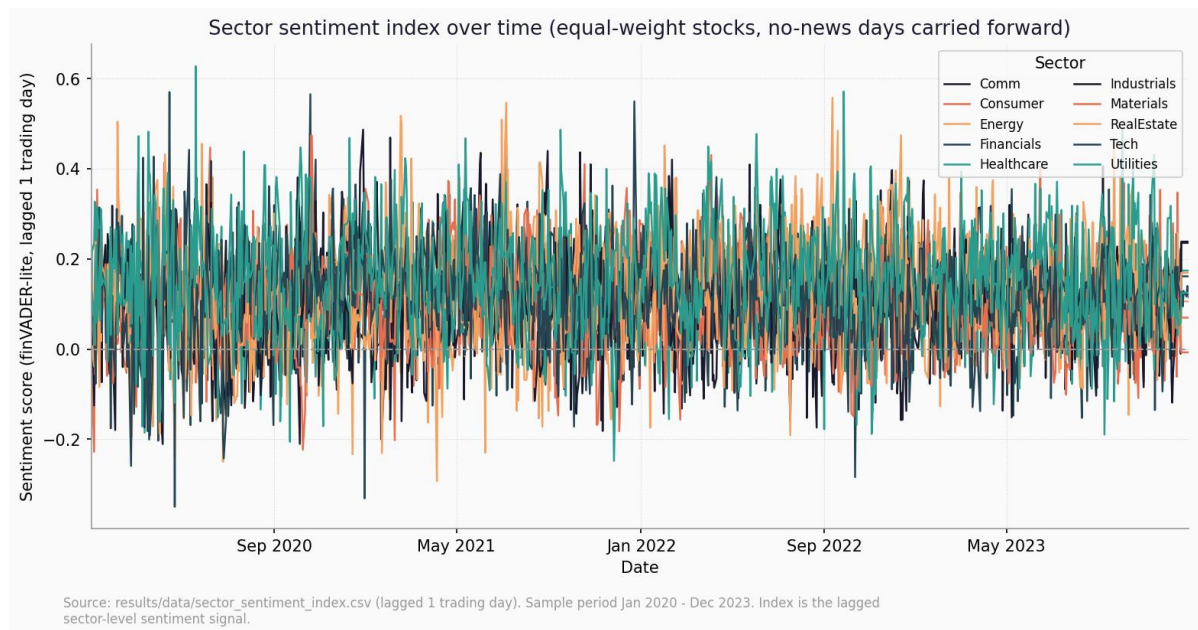


Figure 10: Daily sector sentiment index, all 10 equity sectors (equal-weight stocks; no-news days carried forward). Source: results/data/sector_sentiment_index.csv (lagged 1 trading day). Sample period 2 January 2020 – 29 December 2023.

Figure 10 shows the index over time for all ten sectors, and its levels have a clear cross-sectional order. Over the full sample, Utilities carries the highest mean sentiment (0.166), followed by Real Estate (0.134) and Technology (0.127), while Financials sits at the bottom (0.098). The market-wide equal-weighted index stays positive on 99.3% of trading days and ranges from -0.08 to 0.26. The uniformly positive level is a property of the headlines, not a forecast — company news is written to be read, so neutral language is uncommon and the model’s zero-mean prior is shifted up — and the index should be read as a relative signal across sectors and over time, not an absolute mood thermometer.

The index reads as a relative signal partly because of how quiet days are handled. On days a sector has no headline, the previous score is carried forward rather than dropped, so the series stays smooth and never hollows out during news droughts — but the smoothing means the level partly reflects the recency of coverage, not only its tone. The practical reading for an investor is unchanged: use the index to compare one sector against another, or one month against the next, rather than treating any single level as intrinsically good or bad. This is also the sense in which the index is used downstream: Section 4’s fusion reads each stock’s sentiment relative to its own recent history, not against an absolute threshold.

Two scope statements close this section. Headlines are a noisy proxy for the information investors actually trade on — they are a few words, chosen by editors — so every use of the index carries that caveat. And all sentiment work applies to the equity data only: the cryptocurrency universe has no news dimension in this dataset, so crypto funds are constructed from prices alone, consistent with the brief.

4. Extensions and Innovations

Sentiment fusion: the tilt

The fusion baseline folds the equity sentiment signal into an equity fund through a tilt on top of the base weights. The base is the Invesper US Equity Minimum Volatility Fund, and the tilt rescales each stock’s weight by its recent relative sentiment before renormalising:

$$\tilde{w}_{i,t} = w_{i,t} (1 + \lambda z_{i,t}) / \sum_j w_{j,t} (1 + \lambda z_{j,t}) \quad (4)$$

where $w_{i,t}$ is the base weight of stock i at rebalance t , $z_{i,t}$ is the rolling z-score of stock i ’s lagged sentiment over the past 60 trading days (standardised so the tilt is comparable across stocks), and λ is the tilt direction: $\lambda = +1$ follows sentiment (momentum) and $\lambda = -1$ fights it (contrarian). Negative tilted weights are clipped to zero and the weights renormalised to sum to one, so the tilted fund stays long-only and fully invested. The tilts

are deliberately untuned — no backtest was used to choose λ — and are reported before transaction costs, on the same terms as the base fund.

Table 2: Sentiment fusion before vs after — base equity minimum-variance fund (base), momentum tilt ($\lambda = +1$), contrarian tilt ($\lambda = -1$). Before transaction costs; turnover is average monthly portfolio turnover. Source: results/tables/fusion_comparison.csv.

Fund	λ	Ann. return %	Ann. vol %	Sharpe	Max drawdown %	Total return %	Turnover %
Invesper US Equity Minimum Volatility Fund	0	7.6%	12.8%	0.60	-18.3%	22.5%	7.1%
Invesper US Equity Minimum Volatility Fund (Momentum)	+1	8.3%	13.5%	0.62	-17.0%	24.7%	87.6%
Invesper US Equity Minimum Volatility Fund (Contrarian)	-1	5.3%	13.7%	0.39	-21.2%	14.1%	79.8%

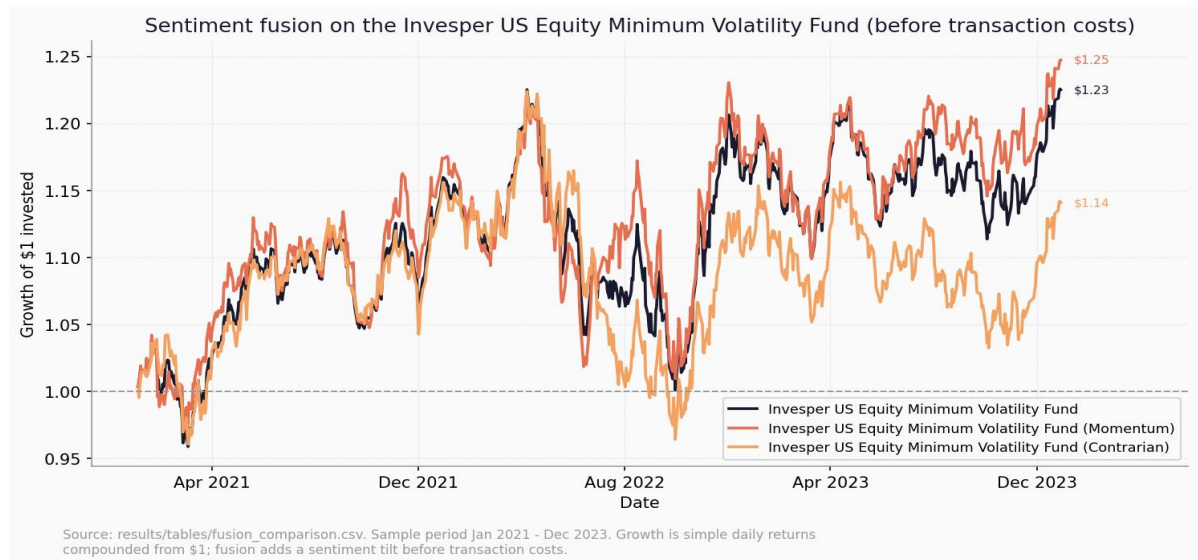


Figure 11: Growth of \$1 — sentiment fusion on the Invesper US Equity Minimum Volatility Fund: base vs momentum vs contrarian (before transaction costs). Source: results/tables/fusion_comparison.csv. Sample period 4 January 2021 – 29 December 2023.

Table 2 and Figure 11 are an honest before-and-after. The momentum tilt improves the base fund's Sharpe from 0.60 to 0.62 and its total return from 22.5% to 24.7%, while the contrarian tilt makes everything worse (0.39 Sharpe, 14.1% total return). The asymmetry is sensible: over 2021–2023 equity sentiment was broadly positive and momentum-

aligned (Section 3), so following it added a little alpha while fighting it removed a lot. But the turnover column is the real story. The base fund turns over 7.1% of its book per month; the momentum tilt turns over 87.6% and the contrarian 79.8% — roughly twelve times more. Under any realistic transaction-cost assumption the tilt’s pre-cost edge is spent several times over in trading costs, which is exactly why Section 1’s zero-cost assumption cannot be silently carried into Section 4, and why a real fund would not run this tilt as-is.

Volatility targeting and the look-ahead bug

The second extension is a volatility-targeting overlay in the spirit of Moreira & Muir (2017): instead of changing the holdings, it scales the fund’s overall exposure up and down so that realised volatility moves toward a fixed target. Applied to the Combined and Crypto Maximum-Sharpe funds, the daily scaling factor is:

$$k_t = \min(\max(\sigma_{\text{target}} / \hat{\sigma}_{t-1}, 0.5), 1.5), \tilde{R}_{p,t} = k_t R_{p,t} \quad (5)$$

where σ_{target} is the target annualised volatility, $\hat{\sigma}_{t-1}$ is the trailing 60-trading-day realised volatility ending strictly before t , and $\tilde{R}_{p,t}$ is the scaled daily return. The clip keeps the fund between 50% and 150% exposure, and the trailing window is causal from the first live day.

This extension contains the project’s most instructive mistake, and it is reported deliberately rather than buried. In the first version, σ_{target} was computed as the fund’s full-sample realised volatility over the entire out-of-sample period — all of 2021–2023. That number only exists after the whole future, including the 2022 crash, has already happened, so on a calm early-2021 day the future-inflated target sat above the trailing realised volatility and the overlay scaled exposure up just before the crash it could not have known about. That is look-ahead bias, the same class of error the rest of the build is asserted against, and it made the Combined fund’s maximum drawdown look worse (-52.7% to -59.3%) rather than better. The -59.3% is the buggy run’s recorded output and cannot be reproduced from the committed CSVs, which the corrected run overwrote; it is logged verbatim in `ai/prompt_log.md` (Entry 7). The fix computes the target once, from the fund’s initial estimation window only — the 2020 pre-live year that set the first weights — and a regression test now asserts the target is a pure function of that window and that corrupting any later date cannot move it (`test_vol_target_target_is_initial_window_only`).

Table 3: Volatility-targeting overlay, base vs vol-targeted, Combined and Crypto Maximum-Sharpe funds (causal targets from the initial estimation window). Source: `results/tables/vol_target_comparison.csv`.

Fund	Target vol	Ann. return %	Ann. vol %	Sharpe	Max drawdown %	Total return %
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Invesper Multi-Asset Opportunities Fund	—	14.1%	34.8%	0.40	-52.7%	27.0%
Invesper Multi-Asset Opportunities Fund (Managed Vol)	0.48	12.5%	40.7%	0.31	-67.0%	13.4%
Invesper Digital Assets Opportunities Fund	—	36.5%	79.0%	0.46	-85.4%	16.4%
Invesper Digital Assets Opportunities Fund (Managed Vol)	0.79	37.1%	78.8%	0.47	-88.0%	18.8%

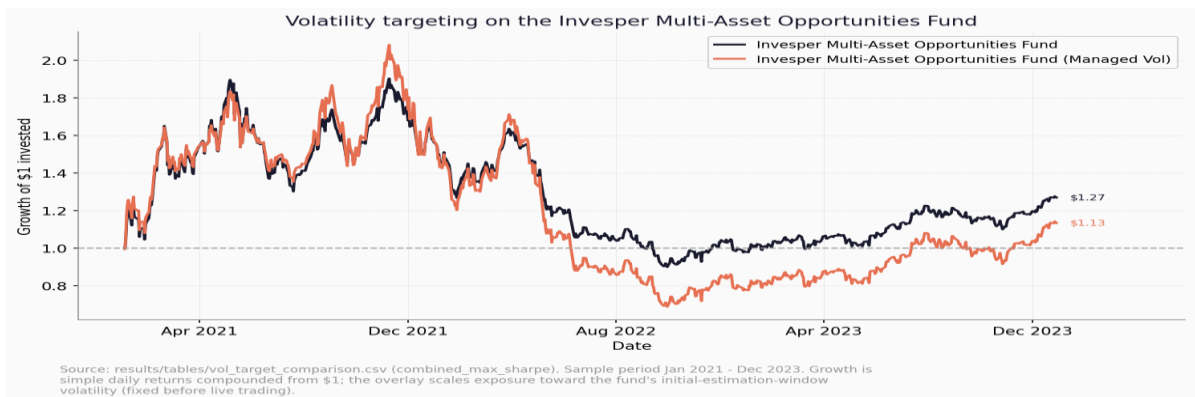


Figure 12: Growth of \$1 — Invesper Multi-Asset Opportunities Fund vs its managed-volatility overlay. Source: results/tables/vol_target_comparison.csv (combined_max_sharpe). The overlay scales exposure toward the fund's initial-estimation-window volatility.

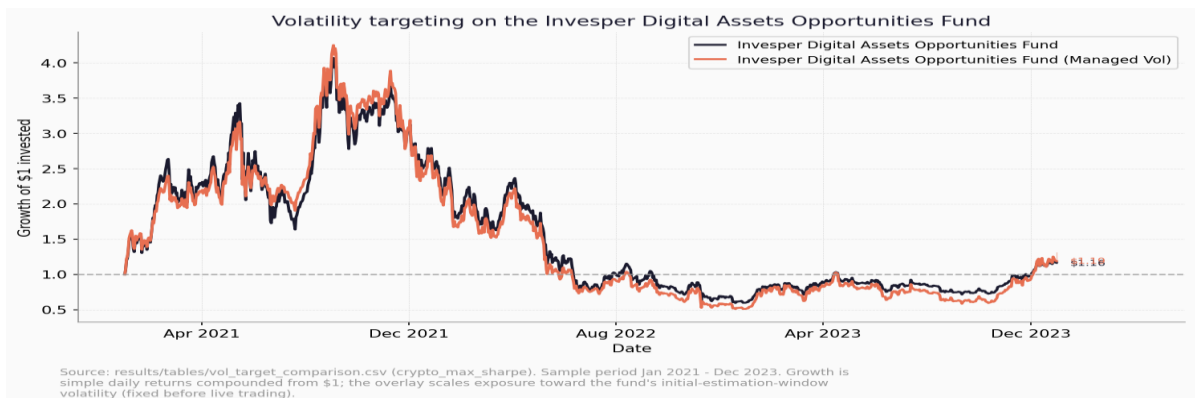


Figure 13: Growth of \$1 — Invesper Digital Assets Opportunities Fund vs its managed-volatility overlay. Source: results/tables/vol_target_comparison.csv (crypto_max_sharpe). The overlay scales exposure toward the fund's initial-estimation-window volatility.

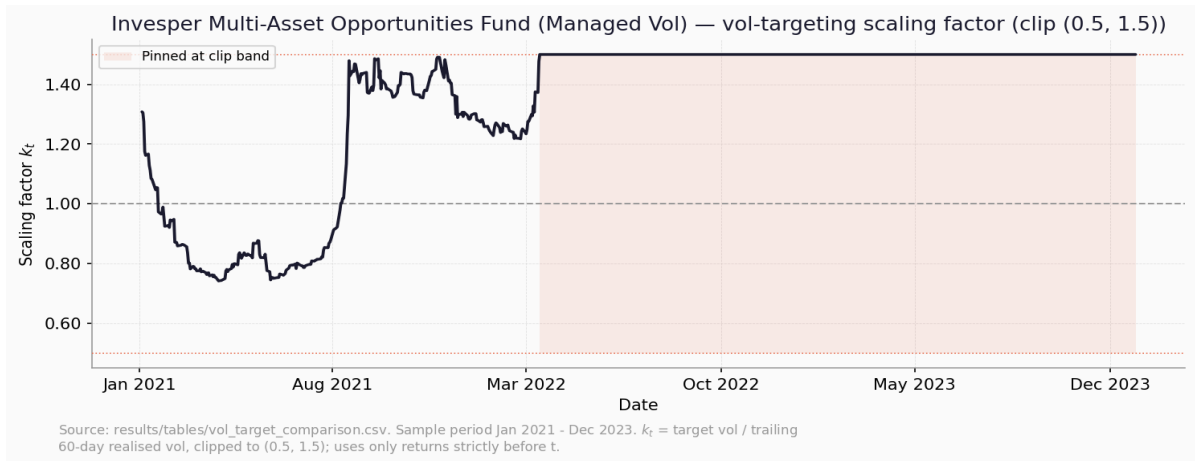


Figure 14: Volatility-targeting scaling factor k_t — Invesper Multi-Asset Opportunities Fund overlay. Source: results/tables/vol_target_comparison.csv. k_t = target vol / trailing 60-day realised vol, clipped to [0.5, 1.5]; uses only returns strictly before t .

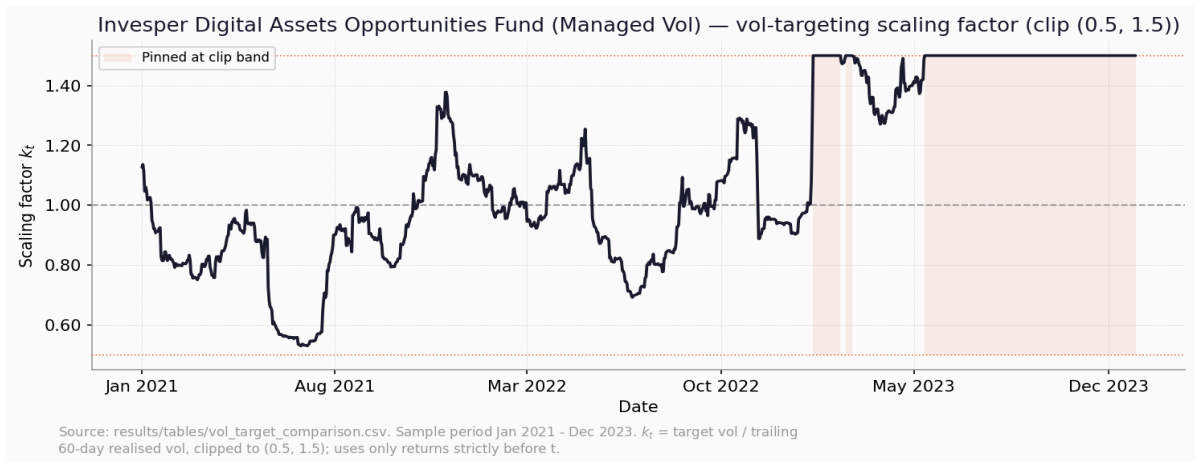


Figure 15: Volatility-targeting scaling factor k_t — Invesper Digital Assets Opportunities Fund overlay. Source: results/tables/vol_target_comparison.csv. k_t = target vol / trailing 60-day realised vol, clipped to [0.5, 1.5]; uses only returns strictly before t .

Table 3 and Figures 12–15 report the corrected, honest result. On the Combined fund the overlay makes things worse: Sharpe falls from 0.40 to 0.31, total return from 27.0% to 13.4%, and maximum drawdown deepens from -52.7% to -67.0%. On the Crypto fund it is a near no-op: Sharpe barely moves (0.46 to 0.47) with a still-worse drawdown (-85.4% to -88.0%). The mechanism is visible in the scaling-factor figures: the 2020 estimation window happens to include the COVID crash, so the target volatility (0.48 for Combined,

0.79 for Crypto) sits well above the 2021–2023 trailing volatility, and k_t exceeds 1 for most of the sample — the overlay spends the whole period leveraging the fund up rather than de-risking it, which is precisely why the 2022 crash hurts more. The brief explicitly credits a careful extension with a negative result, and this one is reported exactly as the corrected numbers show, with nothing tuned to look better.

5. The App and the Investor Journey

The product Invesper actually ships is a Streamlit app deployed from a public GitHub repository (entrypoint `streamlit_app.py` at the repo root). Its design constraint is deliberate: the deployed app reads precomputed artifacts from `results/` only and never runs a backtest, an optimiser, or a sentiment model at request time — the free hosting tier cannot, and the loads are cached so the app stays responsive on a basic machine. All 26 funds across three universes are grouped directly from the data rather than hardcoded: Equity (12 funds), Crypto (9), and Multi-Asset (5), where the fusion variants sit in Equity and the managed-volatility variants inherit their base fund’s universe.

The app is organised around the five actions a user actually takes. Compare shows a per-universe metrics table and a sorted horizontal Sharpe chart, with a radio switch between universes. Fund fact sheet opens any of the 26 funds by its investor-facing name and shows growth of \$1, drawdown, the four headline metrics, and current holdings; the managed-volatility funds show their base fund’s holdings with an explicit caption that the overlay scales overall exposure, not individual holdings. Sentiment analytics shows the sector index lines with multi-select and a market-wide 0–100 fear/greed gauge (0 = fear, 100 = greed), with a caption stating the gauge is unlagged and display-only while fund construction uses the lagged signal. Allocation is the money step: a user builds a portfolio from any of the 26 funds, each slider showing its annual fee, and watches the gross-versus-net growth lines driven by the blended fee. Portfolio is where the allocation becomes a saved (session-only) portfolio.

Table 4: Per-fund annual management-fee schedule, all 26 app-eligible funds.
Source: `results/data/fund_fees.csv`.

Fund	Annual fee %
Invesper US Equity Index Fund	0.05%
Invesper US Equity Minimum Volatility Fund	0.20%
Invesper US Equity Opportunities Fund	0.45%
Invesper US Equity Balanced Risk Fund	0.30%
Invesper Digital Assets Index Fund	0.25%
Invesper Digital Assets Minimum Volatility Fund	0.45%
Invesper Digital Assets Opportunities Fund	0.75%
Invesper Digital Assets Balanced Risk Fund	0.55%
Invesper Multi-Asset Index Fund	0.15%
Invesper Multi-Asset Minimum Volatility Fund	0.30%
Invesper Multi-Asset Opportunities Fund	0.55%
Invesper Multi-Asset Balanced Risk Fund	0.40%

Invesper Defensive Sectors Index Fund	0.20%
Invesper Defensive Sectors Balanced Risk Fund	0.35%
Invesper Cyclical Sectors Index Fund	0.20%
Invesper Cyclical Sectors Balanced Risk Fund	0.35%
Invesper Growth & Innovation Sectors Index Fund	0.25%
Invesper Growth & Innovation Sectors Balanced Risk Fund	0.40%
Invesper Digital Payments Index Fund	0.35%
Invesper Digital Payments Balanced Risk Fund	0.50%
Invesper Web3 Infrastructure Index Fund	0.40%
Invesper Web3 Infrastructure Balanced Risk Fund	0.55%
Invesper US Equity Minimum Volatility Fund (Momentum)	0.85%
Invesper US Equity Minimum Volatility Fund (Contrarian)	0.85%
Invesper Multi-Asset Opportunities Fund (Managed Vol)	0.75%
Invesper Digital Assets Opportunities Fund (Managed Vol)	0.95%

Table 4 is the pricing model. Fees range from 0.05% for the equity index fund to 0.95% for the managed-volatility crypto fund, with the fusion tilts priced at 0.85% to reflect their active turnover. A portfolio's blended fee is the weighted average of its constituents' fees by allocation, charged daily on assets under management, and it drives the gap between the gross and net growth lines a user sees in the Allocation tab.

The zero-commission framing is the customer story behind that fee design. Invesper follows the model the retail brokerage and robo-advisor industry moved to after 2019: no per-trade commission, with revenue coming from a transparent management fee instead. This is what makes Section 1's zero-transaction-cost backtest assumption coherent — it is not a convenient blind spot but the product's stated commercial model — and the app shows the trade-off honestly by plotting gross and net growth side by side rather than hiding the fee line.

The investor journey a user actually follows is: open the app, compare the funds on the metrics table and Sharpe chart to shortlist candidates; open each shortlist fund's fact sheet to check drawdown and current holdings; check the sentiment analytics to see whether a sector is being talked about differently from its price history; set an allocation in the Allocation tab and watch the blended fee reduce the projected growth; and submit the portfolio to see the final weights, growth, and metrics in the Portfolio tab. Every label the user reads is an investor-facing name, not a technical fund id, so the product reads like a fund platform rather than a research output.

6. Critical Reflection and Three Concrete Recommendations

Taken as a whole, the build teaches a consistent lesson: on three years of out-of-sample data, the products that do least win. The equal-weight index funds beat every optimised equity fund (Section 2); risk parity is the strongest combined family; the two deliberate complexity bets — the sentiment tilt and the volatility overlay — either spend their edge on turnover or fail outright (Section 4). For a real product this is a sharper finding than

any individual Sharpe ratio: it says the value of Invesper's shelf is not the optimisation engine but the honest comparison of simple, transparent, low-cost diversification against the alternatives — and that the fee model in Section 5 is the correct one to charge for it, because it aligns revenue with the product that actually works.

The volatility overlay's behaviour is the second lesson. Its result depended on the calibration window in a way that is easy to miss: because the 2020 estimation window includes the COVID crash, the target sits above the fund's live realised volatility and the overlay systematically leverages rather than de-risks. The practical implication is that a volatility target is only as honest as the window it is calibrated on, and a target fixed before a regime change can push a fund in the wrong direction for years. The look-ahead version of this bug — the full-sample target — is the project's most useful AI-workflow artefact, because it is exactly the kind of plausible-sounding, subtly wrong output that a careful reviewer and a causal assertion catch.

However, the analysis and prototypes' scope is limited by the lack of data. As previously mentioned this project uses 50 equities and 10 cryptocurrencies, it can not be used to represent the entirety of the real equities and cryptocurrencies market. Furthermore, sentiment analysis has only been built based on headlines, news contents are needed to create a more robust and accurate sentiment analysis.

Three recommendations follow directly from the evidence. Each is specific enough to act on and is flagged for review because it is a judgement, not a measured fact.

Recommendation 1: price turnover into the fusion before claiming any alpha. The momentum tilt's pre-cost Sharpe gain is 0.02 points against 87.6% monthly turnover, and a real fund would pay for that repeatedly. The next iteration should add a spread-plus-fee transaction-cost model to the backtest and re-report the tilts net of it; if the edge survives, the zero-commission pricing story can be extended honestly to the fused funds, and if it does not, the tilt should be shelved or redesigned with a rebalancing threshold to cut turnover.

Recommendation 2: make the volatility target adaptive rather than fixed to a single pre-live window. The overlay's failure is traceable to a target calibrated on a crisis window, so a rolling or re-estimated causal target — for example a trailing 12-month target recomputed each rebalance from returns strictly before it — would let the overlay de-risk in quiet markets instead of leveraging into a crash. This should be re-tested with the same causal assertions before it is presented as a product feature, and the honest negative result of the current fixed-window version should stay in the fact sheets either way.

Recommendation 3: use the sentiment index where it is reliable and drop it where it is not. The sector-level index is the stronger artefact — it is built from many headlines per day and shows a stable cross-sectional order (Section 3) — yet the fusion only used a

single fund's stock-level z-scores. The next version should fuse sector sentiment into the sector funds instead, or use the market-wide gauge for a defensive cash timing rule, and should keep the honest frame that sentiment is a refinement of the base fund, not a replacement for it.

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